

Base change and automorphic induction: the quadratic rungs on one parity law

Samuel Lavery*

Draft — August 6, 2026

Abstract

Quadratic base change and quadratic automorphic induction, at the level of local Euler data and global coefficient banks, both reduce to one finite combinatorial identity — the parity law $h_j(w \oplus (-w)) = [2 \mid j] \cdot h_{j/2}(w^2)$, difference of squares for complete homogeneous symmetric functions. We prove the parity law at every rank by induction over the transport-closure Cauchy law, derive the split/inert dichotomy of the base-change bank (the two local shapes of the Artin factorization $L(\mathrm{BC}_{E/\mathbb{Q}} \pi, s) = L(\pi, s) L(\pi \otimes \chi, s)$), construct the induced rank-2 bank of a quadratic rank-1 datum with its inert dilation law, and prove the recognition law $\mathrm{AI}(\psi) \otimes \chi = \mathrm{AI}(\psi)$ — the classical fingerprint of induced representations — as a bank identity. An eigenform instance reads off the Frobenius-squared Satake data of $\mathrm{BC}_{E/\mathbb{Q}}(f)$ at inert primes. Everything is unconditional and machine-checked.

1 The parity law

Notation as in the transport-closure note: $h_j(w)$ is the level- j complete homogeneous symmetric function of a finite weight system w , and banks are the multiplicative arithmetic functions assembled from local h -values over prime factorizations. Two preliminary laws, both immediate from the antidiagonal: *homogeneity* $h_j(cw) = c^j h_j(w)$ (each monomial has total degree j ; `radialLocalEulerCoeff_smul`), and the *single channel* $h_j(\alpha) = \alpha^j$ (`radialLocalEulerCoeff_single`).

Theorem 1 (parity law). *For every weight system w of any rank and every level j ,*

$$h_j(w \oplus (-w)) = \begin{cases} h_{j/2}(w^2), & 2 \mid j, \\ 0, & \text{otherwise,} \end{cases}$$

where w^2 is the channel-wise squared system. Equivalently, at the local-factor level: $\prod_i (1 - w_i X)^{-1} \prod_i (1 + w_i X)^{-1} = \prod_i (1 - w_i^2 X^2)^{-1}$.

Proof. Induction on the rank d . Rank 0 is the empty system. Rank 1: by the Cauchy law of the closure note, $h_j(\alpha \oplus (-\alpha)) = \sum_{a+b=j} \alpha^a (-\alpha)^b = \alpha^j \sum_{b \leq j} (-1)^b$, and the alternating sum is 1 for even j , 0 for odd (`parity_rank_one`). Rank $d + 1$: write $w = w' \oplus \alpha$ (one channel split

*Machine-checked companion: `RequestProject/BaseChangeInduction.lean`, 12 named declarations, each at axiom footprint `{propext, Classical.choice, Quot.sound}` or a subset, no `sorry`, no `axiom`, zero `sorryAx`. Every argument below is given in the text; the Lean names locate the corresponding formal statement.

off). Regrouping the doubled system through the relabeling $(w \oplus -w) \simeq (w' \oplus -w') \oplus (\alpha \oplus -\alpha)$ and applying the Cauchy law,

$$h_j(w \oplus (-w)) = \sum_{a+b=j} h_a(w' \oplus (-w')) h_b(\alpha \oplus (-\alpha)).$$

By the inductive hypothesis and rank one, both factors are supported on even levels with half-level values $h_{a/2}(w'^2)$ and $(\alpha^2)^{b/2}$; the even-supported Cauchy sum contracts to the half-level Cauchy sum (`sum_even_pairs`), which by the Cauchy law again is $h_{j/2}(w'^2 \oplus \alpha^2) = h_{j/2}(w^2)$ (`radialLocalEulerCoeff_snoc`). (`parity_law`). $\square$

2 Rung 1: quadratic base change

Fix a quadratic sign χ on primes ($\chi(p) \in \{\pm 1\}$; the character of the quadratic extension $E/\mathbb{Q}$, ramified primes set aside). For a weight-system family w define the *base-change bank*

$$\text{bcBank}(w, \chi) = c_w \star c_{\chi w},$$

the coefficients of $L(\pi, s) L(\pi \otimes \chi, s)$ — classically $L(\text{BC}_{E/\mathbb{Q}} \pi, s)$, the Artin factorization. Its local shape is the split/inert dichotomy:

Theorem 2 (split law). *At a split prime ($\chi(p) = 1$): $\text{bcBank}(p^j) = h_j(w_p \oplus w_p)$ — the doubled system: two places above p , each carrying the original parameter. (`bcBank_split`.)*

Theorem 3 (inert law). *At an inert prime ($\chi(p) = -1$): $\text{bcBank}(p^j) = [2 \mid j] \cdot h_{j/2}(w_p^2)$ — supported on even levels, carrying the Frobenius-squared system: the local factor of the single place above p , with parameter w_p^2 at the norm p^2 . (`bcBank_inert`.)*

Proof. The convolution at p^j is the Cauchy product of the two prime-power sequences. Split: $\chi(p) = 1$ makes the twisted factor the original, and the Cauchy product is $h_j(w_p \oplus w_p)$ by the closure law. Inert: $\chi(p) = -1$ makes the twisted factor $h(-w_p)$, the Cauchy product is $h_j(w_p \oplus (-w_p))$, and Theorem 1 finishes. $\square$

These are exactly the local factors of quadratic base change: the $\mathbb{Q}$ -side product $L(\pi)L(\pi \otimes \chi)$ carries, prime by prime, the E -side Euler data of $\text{BC}_{E/\mathbb{Q}} \pi$.

3 Rung 2: quadratic automorphic induction

A rank-1 quadratic datum consists of split-place pairs $(\gamma_1(p), \gamma_2(p))$ and, at inert primes, a square root $\beta(p)$ of the single place's parameter. The *induced rank-2 system* is

$$\text{AI-weight at } p = \begin{cases} (\gamma_1(p), \gamma_2(p)), & \chi(p) = 1, \\ (\beta(p), -\beta(p)), & \chi(p) = -1, \end{cases}$$

with bank `aiBank` (`aiWeight`, `aiBank`).

Theorem 4 (inert dilation). *At an inert prime, $\text{aiBank}(p^j) = [2 \mid j] \cdot (\beta(p)^2)^{j/2}$: the induced local factor is $(1 - \beta^2 p^{-2s})^{-1}$ — the rank-1 factor of the quadratic place, read at the norm grading. The value depends only on β^2 , and the bank is invariant under the choice of square root $\beta \mapsto -\beta$. (`aiBank_inert`, `aiBank_root_choice`.)*

Theorem 5 (recognition: $\text{AI}(\psi) \otimes \chi = \text{AI}(\psi)$). *Twisting the induced system by the quadratic sign fixes the bank exactly:*

$$c_{\chi \cdot \text{AI-weight}} = \text{aiBank}.$$

Proof. Per prime: at split places $\chi(p) = 1$ and the twist is trivial; at inert places $\chi(p) = -1$ sends $(\beta, -\beta)$ to $(-\beta, \beta)$ — the same multiset, exhibited by the transposition, and the local coefficient is relabeling-invariant. (`aiBank.twist_invariant`.) $\square$

Self-twist invariance is the classical criterion for a $\text{GL}(2)$ representation to be induced from the quadratic extension; here it is a one-line multiset identity — the recognition side of rung 2, complementing the construction side of Theorem 4.

4 The eigenform instance

For a level-one holomorphic Hecke eigenform f with Satake parameters α_p , the seed system $(\alpha_p, \alpha_p^{-1})$ carries the Deligne-normalized coefficient bank of f itself (`seedSystem`, `bankArithmetic_seedSystem` — through the rank-uniform clock-bank identification at $r = 1$). Consequently, at every inert prime of a quadratic extension,

$$\text{bcBank}(\text{seed}, \chi)(p^j) = [2 \mid j] \cdot h_{j/2}(\alpha_p^2, \alpha_p^{-2})$$

(`bcBank_seed_inert`): the local data of $\text{BC}_{E/\mathbb{Q}}(f)$ — the Frobenius-squared Satake pair at the single place above p — read off the $\mathbb{Q}$ -side product $L(f, s)L(f \otimes \chi, s)$, unconditionally.

Remark 6 (register and falsifiers). *Proven and machine-checked: the parity law at every rank, the split/inert dichotomy of the base-change bank, the induced bank’s inert dilation, root-choice invariance, the recognition law, and the eigenform instance. Cited, not re-proved: that these coefficient identities are the local data of the automorphic $\text{BC}_{E/\mathbb{Q}}\pi$ and $\text{AI}(\psi)$ of Langlands and Arthur–Clozel [1] — consumed exactly as local Langlands is consumed elsewhere in this series. Scope: quadratic extensions; the cyclic degree- d case replaces the sign by a d -th root of unity and the parity law by the corresponding d -section law, over the same closure engine. A counterexample would be a weight system and level at which the signed double violates the parity law, or an inert prime at which the base-change bank escapes even support — each formally stated, so a counterexample would require a kernel failure.*

References

- [1] J. Arthur and L. Clozel, *Simple algebras, base change, and the advanced theory of the trace formula*, Ann. of Math. Studies **120**, Princeton (1989).
- [2] R. P. Langlands, *Base change for $\text{GL}(2)$* , Ann. of Math. Studies **96**, Princeton (1980).